

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Shipment Tracking Agent | API-Workflow Domain (Logistics)

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — introduces live external courier tracking API integration

1. Problem Statement

Customer service and procurement teams frequently receive emails asking for the current status of a shipped package — for example, a customer checking when their order will arrive, or a procurement officer confirming whether a vendor's shipment has cleared customs. Today, staff manually copy the tracking number, visit the courier's website, look up the status, and reply by email — a small but repetitive task that delays responses and does not scale during high order volumes such as festive sales periods.

Design and build an Autonomous UiPath Agent, the AI Shipment Tracking Agent, that reads an incoming email requesting a shipment status, extracts the tracking number and courier name, calls a live Courier Tracking API workflow to fetch the current status, and independently replies with the result — without any human approval step.

2. System Prompt

You are an AI Shipment Tracking Agent for the Customer Service and Logistics teams. Your goal is to analyze shipment status request emails and provide an accurate, up-to-date tracking status using a live courier tracking API.

Follow this reasoning process:

1. Read and understand the email content.
2. Classify it into one of the following:
 - Shipment Status Request
 - Delivery Delay Complaint
 - Other
3. If it is Shipment Status Request or Delivery Delay Complaint:
 - Extract the tracking number and courier name (if mentioned)
 - Call the Courier Tracking API workflow with the tracking number
 - Decide if it is urgent (e.g., status shows 'Exception' or 'Delayed')
 - Generate a professional reply stating the current status, last known location, and expected delivery date
4. If it is Other:
 - Skip processing

Rules:

- Think step-by-step before answering

- Be concise and structured
- Do not assume missing details; if the tracking number is missing, ask the customer to provide it instead of guessing
- Do not invent tracking statuses – always use the API response
- Return output in structured format

3. User Prompt

Analyze the following shipment status request email and provide the current tracking status using the live courier tracking API:
{{tracking_email_text}}

4. Input Schema

```
{
  "tracking_email_text": "string"
}
```

5. Output Schema

```
{
  "category": "string",
  "tracking_number": "string",
  "courier_name": "string",
  "current_status": "string",
  "last_location": "string",
  "expected_delivery_date": "string",
  "urgency": "string",
  "action": "string"
}
```

6. Tool: API Workflow & Publish

Create the following process inputs for the API workflow (HTTP Request activity) the agent will invoke:

- tracking_number (text)
- courier_name (text) — optional, used to route to the correct courier API endpoint

Configure the workflow to call a live courier tracking API endpoint (e.g., an HTTPS GET request to a shipment-tracking provider) and parse the JSON response for status, location, and estimated delivery date.

7. Add Tool: Map Inputs

Agent Field	API Request Parameter
tracking_number	tracking_id
courier_name	carrier_code

8. Updated System Prompt — Tool Invocation Logic

If the email is classified as Shipment Status Request or Delivery Delay Complaint:

- Call the `get_shipment_status` API workflow with `tracking_number` and `courier_name` (if available).
- Parse the API response for `'status'`, `'last_location'`, and `'expected_delivery_date'`.
- If the returned status is `'Exception'` or `'Delayed'`, set `urgency = "High"`.
- Use the `generate_reply` GenAI activity to draft a professional reply email with the current status, last known location, and expected delivery date – fully autonomously, with no manual approval required.

If the email is classified as Other:

- Take no further action.

9. Tools & Activities Summary

Tool / Activity	Type
<code>get_shipment_status</code>	API Workflow (HTTP Request to a live courier tracking API)
<code>generate_reply</code>	GenAI Activity (LLM prompt-based text generation)
<code>classify_request</code>	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent classifies the request, calls the live tracking API, evaluates urgency, and composes the reply without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a system prompt that integrates a live external API into agent reasoning
- Configure an API Workflow (HTTP Request activity) with mapped input parameters
- Parse a JSON API response and derive urgency from the returned status
- Define structured input/output schemas for an API-integrated autonomous agent
- Publish and test a fully autonomous, API-driven agent in UiPath Agent Builder